Conversational EDA Chatbot & E-Commerce Customer Churn Prediction

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1. Motivation

Two pain points plague modern e-commerce analytics:

- **Data-access bottleneck.** Stakeholders wait hours or days for analysts to write SQL, killing iterative exploration.
- **Reactive retention.** Churn is discovered *after* revenue loss; acquiring a new customer costs **5–25×** more than retaining one.

Our solution is a unified analytics platform with two artifacts that share the same data domain:

1. A **conversational EDA chatbot** that turns natural language into safe SQL and interactive charts.
2. A **churn-prediction pipeline** that ranks at-risk customers so retention spending is targeted.

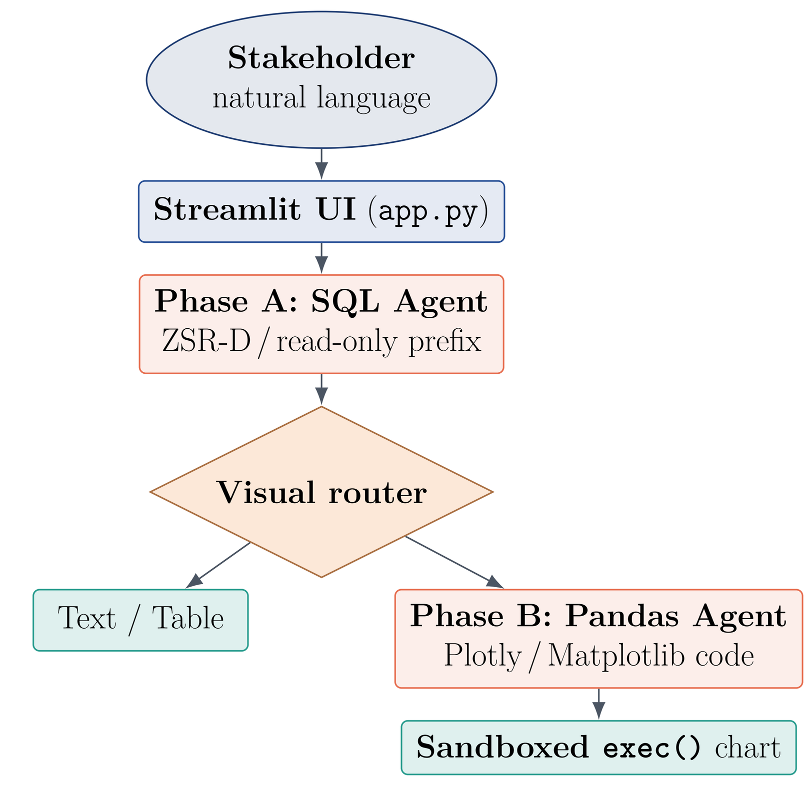
2. Objectives

- Translate plain English → syntactically valid MySQL.
- Dynamic output: **table**, **summary**, or **interactive chart**.
- Strict **read-only** guardrails.
- Multi-turn **conversational memory**.
- Train and evaluate a churn classifier with leakage-free cross-validation, hyperparameter tuning, and interpretability.
- Ship **portable, modular** artifacts (one .py + one .ipynb).

3. Technology Stack

Layer	Technology
Chat UI	Streamlit
LLM orchestration	LangChain (+Experimental)
Language model	OpenAI GPT-4o, $T = 0$
Database	MySQL via `pymysql`
Visualization	Plotly Express, Matplotlib
ML Modeling	scikit-learn
Notebook	Jupyter
Runtime	Python 3.11

4. System Architecture

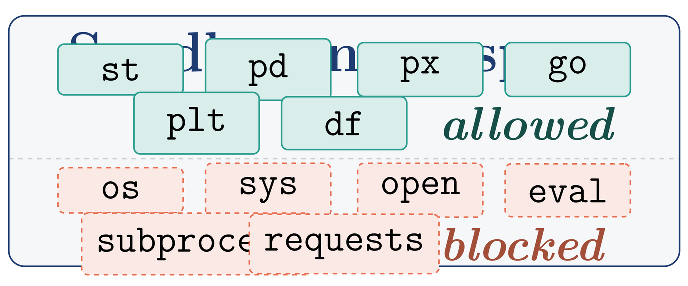


Dual-agent design. Phase A generates SQL; Phase B generates visualization code only when the prompt contains a visual trigger.

5. Phase A — SQL Agent

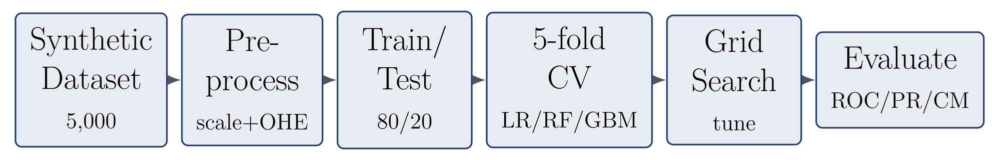
- Built with `create_sql_agent` using the **ZERO_SHOT_REACT_DESCRIPTION** agent type.
- Executes a *Thought* → *Action* → *Observation* loop over the `SQLDatabaseToolkit` (list tables, inspect schema, run query, check query).
- **Read-only system prefix** explicitly forbids `INSERT, UPDATE, DELETE, DROP, TRUNCATE, ALTER, CREATE, REPLACE, GRANT, REVOKE, RENAME, LOCK, CALL, MERGE`.
- Hardened: `handle_parsing_errors=True`, `max_iterations=12`, `sample_rows_in_table_info=3`.
- Connection pooled via `@st.cache_resource`.

6. Phase B — Visualization Sandbox



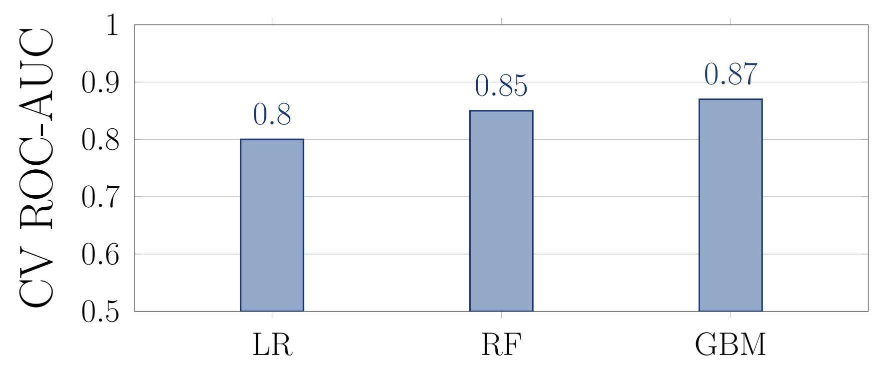
Three-layer defense. (1) prompt-level restriction, (2) namespace-level restriction (only six names in scope), (3) broad `try/except` so the Streamlit thread never crashes.

7. ML Pipeline



Reproducible 5,000-customer e-commerce dataset generated with `seed=42`. Eleven features span demographics (age, gender, region), account (tier, payment, tenure), and behavior (AOV, cadence, return rate, support tickets, recency, discount usage).

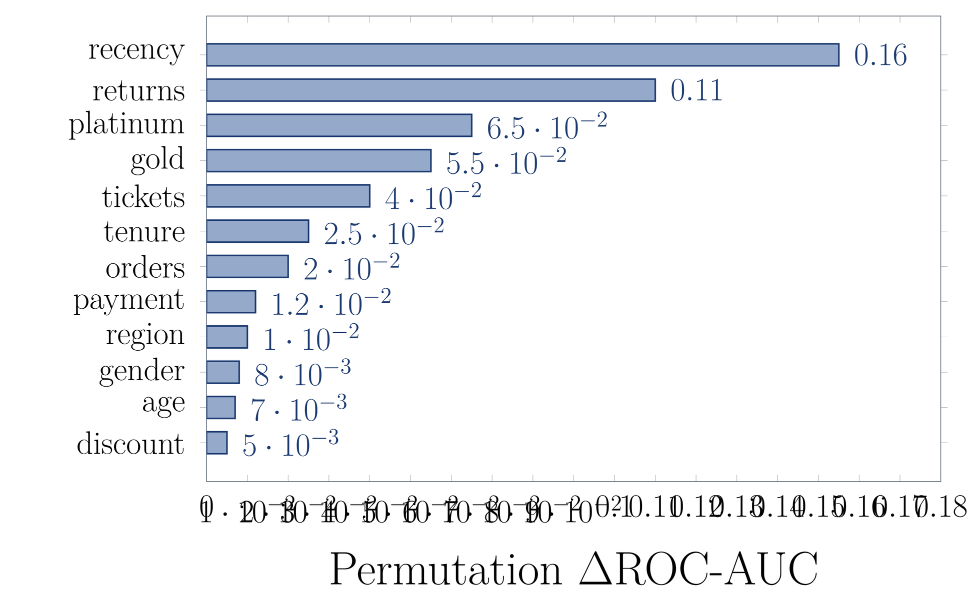
8. Results



Model	CV ROC-AUC	Std
Gradient Boosting	**0.87**	0.010
Random Forest	0.85	0.011
Logistic Regression	0.80	0.012

Tuned Gradient Boosting wins; ensembles beat the linear baseline by a clear margin on the held-out test set.

9. Top Churn Drivers



Recency and **return rate** dominate; **loyalty tiers** (Gold, Platinum) protect against churn.

10. Conclusion

We deliver a coherent two-part analytics solution: a **secure conversational EDA chatbot** that turns plain English into SQL queries and interactive charts, and a **rigorous churn-prediction pipeline** that ranks customers by retention risk. Both components ship with production discipline: caching, error handling, security guardrails, reproducibility, and clean modular code. Together they cover the **descriptive** and **predictive** halves of customer analytics in one platform.

11. Future Work & Acknowledgments

Future work.

- Replace the keyword router with an LLM-as-judge intent classifier.
- Retrain on the live MySQL warehouse via the same connection.
- Per-customer SHAP attributions for personalized retention offers.
- Cost-sensitive threshold tuning + LangSmith observability.

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